

Which of the following compounds can participate in hydrogen bonding with methanol (CH_3OH)?

I. NH_3

II. HF

III. CH_4

☐ A. I only [9%]

✓ ☒ B. I and II only [81%]

☐ C. II and III only [1%]

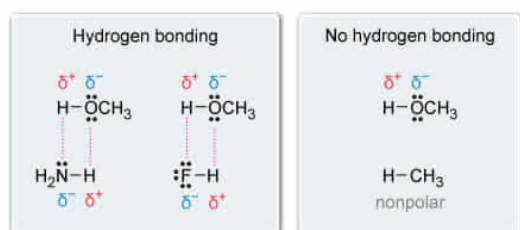
☐ D. I, II, and III [7%]

Correct

82% answered correctly

Explanation:

Hydrogen bonding interactions with methanol



Compounds with polar N-H, O-H, and F-H bonds participate in noncovalent hydrogen bonding interactions

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Hydrogen atoms covalently **bonded with N, O, or F atoms** form **polar bonds** that yield **dipoles** with partial charges. Attractions between opposite partial charges of the dipoles generate a significant intermolecular force known as **hydrogen bonding**. This term does not refer to the covalent chemical bond between hydrogen and another atom but instead refers to the **noncovalent attractions** between the bond dipoles.

Because of the significant difference in **electronegativity** between hydrogen and oxygen, O-H bonds exhibit a strong dipole. The lone pairs of electrons and partial negative charge (δ^-) on the oxygen atom serve as a **hydrogen bond acceptor**, and the partially positive (δ^+) hydrogen atom serves as the donor. As a result, water and alcohols (such as methanol) interact with other compounds that can either accept or donate hydrogen bonding interactions.

Although smaller in magnitude than that of a hydroxyl group, N-H bonds have a significant dipole, and the nitrogen atom possesses a lone pair of electrons. As a result, ammonia and many organic amines readily participate in hydrogen bonding with alcohols (**Number I**).

Likewise, the H-F bond is highly polar, and the fluorine atom has three pairs of nonbonding electrons. Accordingly, highly polar bonds involving fluorine can act as hydrogen bond acceptors with alcohols (**Number II**).

(Number III) The carbon and hydrogen atoms in a C–H bond do not have a sufficient difference in electronegativity to generate a dipole that is capable of hydrogen bonding.

Educational objective:

Hydrogen atoms covalently bonded with nitrogen, oxygen, or fluorine atoms form polar bonds that yield dipoles, which exhibit significant noncovalent dipole-dipole attractions known as hydrogen bonding.

Time spent:0 sec

Subject:
General Chemistry

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System:
Interactions of Chemical Substances